

MPI Parallelisation of Molecular Dynamics Algorithms

BCN: 4316J

Abstract. We present a C++17/MPI molecular dynamics solver for harmonic-oscillator and Lennard–Jones systems, implementing Forward Euler, classical RK4, and Velocity–Verlet within a particle-decomposed parallel framework. Harmonic-oscillator tests recover the expected first-, second-, and fourth-order global convergence of Euler, Velocity–Verlet, and RK4, respectively, while also showing that formal order alone does not determine production suitability: Euler exhibits secular energy growth and phase-space distortion, RK4 achieves the smallest short-time state error, and Velocity–Verlet retains the bounded energy behaviour required for stable long-time Hamiltonian simulation. In a 100-step, 1 ps NVE argon run ($N = 864$, $\rho = 0.8\sigma^{-3}$, $T = 94.4$ K), Velocity–Verlet maintains mean temperature 94.42 K and maximum relative energy deviation 0.082%, whereas Euler heats to 396.05 K with 127.6% energy drift. An extended Velocity–Verlet run yields a radial distribution function in agreement with Rahman’s liquid-argon shell structure. Strong scaling at fixed $N = 2048$ reaches $S(32) = 24.12$ with 75.4% efficiency; fixed- P problem-size scaling follows $T_{\text{wall}} \propto N^{1.85}$, consistent with direct $\mathcal{O}(N^2)$ force evaluation. The particle-decomposition MPI design is effective at moderate core counts but ultimately limited by global communication.

1 Introduction

Classical molecular dynamics (MD) generates trajectories of interacting atoms by numerically integrating their equations of motion under a prescribed potential. It provides a practical route from microscopic dynamics to structural, dynamical, and thermodynamic observables such as radial distribution functions, time-correlation functions, and transport coefficients, while retaining the underlying time evolution itself. This is important because many physically relevant phenomena, including relaxation, metastability, and transport, depend not only on equilibrium structure but on how the system evolves in time. Rahman’s 1964 simulation of liquid argon [5] showed that even a comparatively simple interatomic model, when evolved directly in time, could recover non-trivial liquid structure and dynamics.

For the conservative N -particle systems considered here, the motion is governed by

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i = -\nabla_i V(\{\mathbf{r}_j\}), \quad (1)$$

where the force on particle i depends on the positions of all particles through the potential energy V . This makes MD demanding in two respects. The timestep must be small enough to resolve the fastest motions accurately, yet the total simulated time must still be long enough to extract useful observables from equilibration and sampling. As a result, MD requires both repeated force evaluation within each timestep and very large numbers of timesteps overall.

The same dynamics may equivalently be written in Hamiltonian form,

$$H(\mathbf{r}, \mathbf{p}) = \sum_{i=1}^N \frac{|\mathbf{p}_i|^2}{2m_i} + V(\mathbf{r}_1, \dots, \mathbf{r}_N), \quad (2)$$

which makes clear that an MD code approximates the long-time flow of a Hamiltonian system in phase space. For conservative dynamics, formal truncation-error order alone is therefore not an adequate criterion for an integrator: long-time physical fidelity depends on whether the discrete update respects time-reversibility

and the symplectic structure of the underlying flow. Forward Euler is straightforward but neither time-reversible nor symplectic, so it distorts phase-space volume and produces systematic energy drift. Velocity–Verlet is both symplectic and time-reversible, giving the bounded long-time energy behaviour required for stable MD. Classical fourth-order Runge–Kutta (RK4) attains smaller short-time truncation error but does not share these geometric guarantees; in a particle-decomposition MPI setting it is also operationally expensive, because each of its four intermediate force evaluations requires a globally consistent particle configuration. For molecular dynamics, where long-time statistical behaviour matters as much as local accuracy, these distinctions are fundamental.

These numerical questions are examined here using two complementary model problems:

$$V_{\text{HO}}(x) = \frac{1}{2}kx^2, \quad V_{\text{LJ}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]. \quad (3)$$

The harmonic oscillator provides an analytically solvable benchmark for convergence order, timestep sensitivity, and phase-space behaviour, with no ambiguity from model error. The Lennard–Jones potential then moves to a standard interacting many-particle system, following Rahman’s liquid-argon study with $N = 864$, fixed density, and standard parameters. Comparison with Rahman is interpreted as a check on features of an equilibrated Lennard–Jones liquid under comparable thermodynamic conditions, rather than as an attempt at exact trajectory-level agreement.

A further challenge is computational. With direct pairwise force evaluation, the dominant work scales as $\mathcal{O}(N^2)$ per timestep, so serial execution rapidly becomes restrictive as either system size or simulation length grows. Parallelisation is therefore integral to modern MD, particularly on current CPU architectures where clock speeds have largely plateaued while core counts continue to rise. In the present direct-summation, particle-decomposition code, distributing particles across P MPI ranks reduces the force-evaluation cost towards $\mathcal{O}(N^2/P)$, but the force

on each local particle still depends on positions that may reside on other ranks, so communication is unavoidable. A simple performance model is

$$t_P \approx t_s + \frac{t_p}{P} + t_{\text{comm}}(P), \quad (4)$$

where t_s denotes serial work, t_p the ideally parallel work, and $t_{\text{comm}}(P)$ the communication overhead. Scaling is governed not only by arithmetic work but also by the cost of maintaining globally consistent particle data. The present implementation uses particle decomposition because it gives a transparent parallelisation of direct-force MD; spatial domain decomposition, in which communication is largely restricted to neighbouring subregions, is the standard production alternative and is discussed further in the conclusions.

This report examines a C++17/MPI solver implementing Forward Euler, RK4, and Velocity-Verlet. The harmonic oscillator verifies convergence behaviour and distinguishes formal order from long-time structure preservation; the Lennard–Jones argon system tests physical plausibility and numerical stability against Rahman’s classical results; and scaling experiments quantify the gain from distributing $\mathcal{O}(N^2)$ force work against the communication cost of maintaining consistent particle data across ranks. The aim is to assess whether the numerical methods, physical model, and parallel design are jointly appropriate for molecular dynamics on modern distributed-memory systems.

2 Methodology

2.1 Governing equations and time integration

For a system of particles with positions $\mathbf{r}_i$, velocities $\mathbf{v}_i$, masses m_i , and potential energy $V(\{\mathbf{r}_j\})$, the equations of motion are

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i(\{\mathbf{r}_j\}), \quad \mathbf{F}_i = -\nabla_i V. \quad (5)$$

Writing the dynamics as a first-order system for the state vector $\mathbf{y} = (\mathbf{r}, \mathbf{v})$ gives

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \quad \dot{\mathbf{v}}_i = \mathbf{a}_i(\mathbf{r}) = \mathbf{F}_i(\mathbf{r})/m_i. \quad (6)$$

Three explicit time integrators were implemented: Forward Euler, classical fourth-order Runge–Kutta (RK4), and Velocity-Verlet.

2.1.1 Forward Euler

Forward Euler advances positions and velocities by first-order Taylor truncation,

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \Delta t \mathbf{v}_n, \quad (7)$$

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \Delta t \mathbf{a}_n. \quad (8)$$

Its local truncation error is $\mathcal{O}(\Delta t^2)$ and its global error over a fixed time interval is $\mathcal{O}(\Delta t)$. Euler is simple and useful as a baseline, but it is neither time-reversible nor symplectic. For Hamiltonian dynamics this typically leads to phase-space distortion and systematic energy drift.

2.1.2 Classical RK4

RK4 applies the classical four-stage explicit Runge–Kutta method to Eq. (6),

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(\mathbf{y}_n), \\ \mathbf{k}_2 &= \mathbf{f}\left(\mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_1\right), \\ \mathbf{k}_3 &= \mathbf{f}\left(\mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_2\right), \\ \mathbf{k}_4 &= \mathbf{f}(\mathbf{y}_n + \Delta t \mathbf{k}_3), \\ \mathbf{y}_{n+1} &= \mathbf{y}_n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned} \quad (9)$$

Because the state vector $\mathbf{y}$ couples position and velocity, each intermediate stage requires an intermediate force evaluation. RK4 has local truncation error $\mathcal{O}(\Delta t^5)$ and global error $\mathcal{O}(\Delta t^4)$, so it is a useful high-order reference on the harmonic oscillator. It is not symplectic, however, so small short-time state error does not guarantee good long-time thermodynamic fidelity. In the MPI LJ implementation, each stage requires a globally consistent intermediate position field, so RK4 incurs four `MPI_Allgatherv` calls per timestep. Since one global position reconstruction transfers $3N$ doubles, RK4 communicates $12N$ doubles per timestep, four times the position-exchange volume of Euler or Velocity-Verlet.

2.1.3 Velocity-Verlet

Velocity-Verlet is implemented in half-kick / drift / force / half-kick form,

$$\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{\Delta t}{2} \mathbf{a}_n, \quad (10)$$

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \Delta t \mathbf{v}_{n+1/2}, \quad (11)$$

$$\mathbf{a}_{n+1} = \mathbf{F}(\mathbf{r}_{n+1})/m, \quad (12)$$

$$\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{\Delta t}{2} \mathbf{a}_{n+1}. \quad (13)$$

For the coupled $(\mathbf{r}, \mathbf{v})$ update, the local truncation error is $\mathcal{O}(\Delta t^3)$ and the global trajectory error is $\mathcal{O}(\Delta t^2)$. Although formally lower-order than RK4, Velocity-Verlet is time-reversible and symplectic for conservative dynamics, so one expects bounded oscillatory energy error for sufficiently small Δt . It also requires only one new force evaluation and one global communication phase per timestep, making it the production integrator used for the LJ system.

The harmonic oscillator is used first as an analytically solvable validation problem, before moving to the interacting many-particle Lennard–Jones system.

2.2 Interaction model and force evaluation

The interacting-particle test case is a Lennard–Jones (LJ) argon fluid at Rahman’s liquid-argon state point. Pair interactions are defined by

$$V(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right], \quad (14)$$

with argon parameters $\varepsilon/k_B = 120$ K, $\sigma = 3.4$ Å, and $m_{\text{Ar}} = 66.904265 \times 10^{-27}$ kg. The corresponding accel-

eration on particle i is computed by direct pair summation,

$$\mathbf{a}_i = \frac{24\epsilon}{m} \sum_{j \neq i} \frac{\mathbf{r}_{ij}}{r_{ij}^2} \left[2 \left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right], \quad (15)$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$.

A hard cutoff $r_c = 2.25\sigma$ is applied, with no shifted-potential or shifted-force correction, so the interaction is set to zero for $r > r_c$. This introduces small discontinuities when pairs cross the cutoff. The simulation domain is a periodic cubic box. For $N = 864$ particles the side length is $L = 10.229\sigma$, and for other N it is scaled as $L = 10.229\sigma(N/864)^{1/3}$ to maintain constant number density $\rho = 0.8\sigma^{-3}$. Periodicity is handled through the minimum image convention,

$$\Delta x \leftarrow \Delta x - L \text{round}(\Delta x/L), \quad (16)$$

and analogously in y and z . In the implementation, the division is replaced by multiplication with a precomputed inverse box length. The condition $r_c < L/2$ is satisfied for the Rahman state point.

The code operates directly in SI units, with state, constants, and arithmetic stored in C++ `double` (IEEE-754 binary64). Using direct SI units combines quantities on very different scales, such as distances of $\mathcal{O}(10^{-10})$ m and masses of $\mathcal{O}(10^{-27})$ kg, so binary64 precision was used throughout to reduce loss of significance and maintain stable force, energy, and convergence measurements in the present tests. Forces are evaluated by a direct all-pairs $\mathcal{O}(N^2)$ loop rather than by neighbour lists or linked-cell methods, keeping the force kernel simple and the scaling behaviour transparent. Derived LJ factors are precomputed once, and within the hot loop the implementation avoids `std::pow` and `std::sqrt`: it forms r^{-2} and then chains multiplications to obtain r^{-6} and r^{-12} . A small numerical guard of order $10^{-30}\sigma^2$ is applied to the squared separation before inversion to avoid overflow or division by an effectively zero separation during initialisation or early relaxation.

Potential energy is accumulated over all $j \neq i$ interactions on each rank and halved after reduction, avoiding a branch in the inner pair loop at the cost of evaluating each pair twice. Newton's third law is not exploited across ranks. Under the contiguous block decomposition used here, a cross-rank half-list based on $i < j$ would give low-index ranks more pair work than high-index ranks. Full local-against-global force evaluation therefore trades additional arithmetic for balanced work distribution and a simpler communication pattern.

2.3 Parallel decomposition and communication mechanics

The solver uses particle decomposition. At startup, the global particle index range $1 \dots N$ is partitioned into fixed contiguous blocks across P MPI ranks. When N is not an exact multiple of P , all ranks receive $\lfloor N/P \rfloor$ particles and the first $N \bmod P$ ranks receive one additional particle. The local particle count on rank r is

therefore

$$N_r^{\text{loc}} = \left\lfloor \frac{N}{P} \right\rfloor + (r < N \bmod P). \quad (17)$$

Each rank retains ownership of its block for the duration of the simulation. This distributes the $\mathcal{O}(N^2)$ force evaluation, but the local force on any owned particle still depends on the positions of all N particles.

Global position synchronisation is handled with `MPI_Allgatherv` at the start of each force-evaluation step. Because N_r^{loc} varies across ranks, `MPI_Allgather` is not sufficient. The receive counts (`recvcounts`) and displacements (`displs`) are computed during initialisation. For 3D coordinates, `recvcounts[k] = 3N_k^{\text{loc}}`, and

$$\text{displs}[k] = \text{displs}[k-1] + \text{recvcounts}[k-1].$$

This reconstructs a contiguous global position buffer on every rank.

For Velocity-Verlet and Euler, `MPI_Allgatherv` is called once per timestep. RK4 requires four intermediate global position reconstructions per timestep, increasing the communication volume by a factor of four. The harmonic-oscillator mode bypasses this inner-loop communication because the particles are non-interacting, which isolates truncation error from MPI overhead.

Other MPI collectives are used for setup and measurement. Initial particle data are generated on rank 0 and distributed with `MPI_Bcast`, so runs at different process counts begin from identical initial conditions. Aggregate observables such as kinetic and potential energy are combined with `MPI_Reduce(MPI_SUM)`, and wall time is reported from the slowest rank using `MPI_Reduce(MPI_MAX)`.

2.4 Architectural considerations

Two implementation choices are relevant to the measured performance. First, Newton's third law is not exploited across rank boundaries. Although this would reduce arithmetic work in principle, it would also require an additional communication phase to return partial force contributions to the owning ranks. The present implementation instead evaluates the full local-against-global interaction set, favouring balanced work and a simpler communication pattern.

Second, particle state is stored in an Array of Structures (AoS) layout, with coordinates flattened as `arr[3*i+d]`. This gives a contiguous MPI buffer and preserves locality for each 3D pair-distance calculation. A Structure of Arrays (SoA) layout would generally be more SIMD-friendly for the force kernel, so the present layout is convenient but not maximally vectorisation-friendly.

2.5 Initialisation, equilibration, and reproducibility

For the LJ argon system, particles are initialised on a face-centred cubic lattice at the required density. To break exact symmetry in the initial force evaluation,

each coordinate is perturbed by a zero-mean uniform displacement of magnitude $\pm 0.01\sigma$. Initial velocities are generated on rank 0 via a Box–Muller transform using a fixed seed with the 64-bit Mersenne Twister engine `std::mt19937_64`, and the resulting global state is broadcast so that all runs start from the same microscopic configuration.

After velocity generation, the centre-of-mass velocity is removed by subtracting the mean velocity in each Cartesian direction. This enforces zero total momentum and prevents bulk drift through the periodic box. Temperature is therefore evaluated using

$$N_{\text{dof}} = 3(N - 1)$$

degrees of freedom rather than $3N$.

The target temperature is 94.4 K. Because the perturbed FCC lattice is not an equilibrium liquid configuration, a 50-step startup phase is used. During this stage, velocities are uniformly rescaled each step according to

$$\lambda = \sqrt{T_{\text{target}}/T},$$

with the rescaling factor broadcast to all ranks. Unlike extended-system or stochastic thermostats, this is simply a short startup procedure to bring the system close to the Rahman state point and reduce the initial transient. Because it does not preserve Hamiltonian phase-space volume, it is not used for reporting observables. A final exact rescale is applied at the boundary between startup and production, after which the trajectory is propagated in unthermostatted NVE dynamics so that the reported sampling comes from the Hamiltonian flow.

Reproducibility is enforced by generating random numbers centrally with a fixed seed and then broadcasting the resulting state, so physical and timing comparisons across process counts are like-for-like.

2.6 Validation and performance metrics

The harmonic oscillator provides an exact reference problem,

$$\begin{aligned} \ddot{x} &= -\omega^2 x, & x(0) &= 1, & \dot{x}(0) &= 0, \\ x_{\text{exact}}(t) &= \cos t & (\omega &= 1), \end{aligned} \quad (18)$$

against which numerical error can be measured without model uncertainty. Global convergence is assessed using the endpoint position error

$$\epsilon_{\text{end}}(\Delta t) = |x_{\text{num}}(T) - x_{\text{exact}}(T)|, \quad T = 10, \quad (19)$$

together with an RMS phase-space error over the trajectory. Phase-space portraits and relative energy deviation are also used to distinguish formal order from long-time geometric fidelity.

For the LJ system, a 100-step NVE production run at $\Delta t = 10^{-14}$ s is used to satisfy the required 10^{-12} s simulation duration, assessing energy conservation and temperature stability after startup. Structural validation is supplemented by a separate extended Velocity-Verlet run of 20 000 production steps, with the radial

distribution function accumulated after a 200-step discard and sampled every five steps; the required 100-step run is too short for a meaningful RDF. Because the RDF is constructed from unordered pairs, its normalisation uses the finite-size ideal-gas density $(N - 1)/V$ rather than N/V .

Parallel correctness is checked by comparing runs with identical initial conditions across process counts. Agreement between $P = 1$ and $P = 2$ is bitwise, while any tiny differences at larger P arise from floating-point non-associativity in reduction order rather than from different underlying dynamics.

All timing runs are performed without file output. Let T_P denote the maximum wall time across ranks for a run on P processes. Strong-scaling performance is quantified by

$$S(P) = \frac{T_1}{T_P}, \quad E(P) = \frac{S(P)}{P}, \quad (20)$$

and communication time is measured as the maximum accumulated `MPI.Allgatherv` time across ranks. The strong-scaling curve is also compared against Amdahl's law,

$$S(P) = \frac{1}{f + (1 - f)/P}, \quad (21)$$

where f is used as an empirical descriptor of the measured curve rather than as a literal machine-independent serial fraction.

3 Results

3.1 Harmonic Oscillator Verification

The harmonic oscillator ($\omega = 1$, $x(0) = 1$, $\dot{x}(0) = 0$, $T = 10$) isolates the numerical properties of the three integrators on a problem whose exact solution is known, so that all differences between schemes are attributable to the numerics rather than to model uncertainty.

At $\Delta t = 0.01$, the position and velocity traces in Fig. 1(a,b) remain close to the exact solution for all three schemes over $0 \leq t \leq 10$; on this scale the overlays serve mainly as an implementation check. The phase-space portrait in Fig. 1(c) is more discriminating: Euler spirals outward from the exact closed orbit, consistent with non-symplectic phase-space distortion; Velocity-Verlet remains near a closed curve; and RK4 is visually indistinguishable from the exact orbit. Phase-space geometry therefore separates the integrators more effectively than time-trace overlays, and already illustrates the distinction between formal accuracy and structural preservation described in the Methodology.

Figure 2 contrasts a fine timestep ($\Delta t = 0.01$) with a coarse one ($\Delta t = 0.5$) to separate formal order from structural robustness. At the coarse step, Euler becomes qualitatively unreliable: the amplitude grows rapidly and the phase-space orbit expands outward. Velocity-Verlet at the same step is no longer quantitatively accurate but remains bounded and preserves the oscillatory character. RK4 remains close to the exact solution. High formal order (RK4) and structural ro-

bustness (Velocity-Verlet) are thus distinct properties; Euler possesses neither.

Formal convergence rates are confirmed quantitatively by Fig. 3. Over the fitted range $\Delta t \leq 0.1$, the endpoint position error at $T = 10$ yields slopes of 1.05, 2.00, and 3.94 for Euler, Velocity-Verlet, and RK4, while the RMS phase-space error gives 1.03, 2.00, and 4.00. These recover the expected first-, second-, and fourth-order global convergence (Eqs. 7–8, 10–13, and 9). Representative endpoint errors spanning five orders of magnitude are listed in Appendix Table 3. At the finest tested step ($\Delta t = 5 \times 10^{-4}$), the RK4 endpoint error reaches 3.6×10^{-15} , consistent with the IEEE 754 double-precision floor; further refinement beyond this point is limited by machine arithmetic rather than by the integrator.

The supporting energy diagnostic in Appendix Fig. 9 complements the convergence data. At $\Delta t = 0.01$, Euler exhibits secular energy drift (max relative deviation 10.5%), Velocity-Verlet shows bounded oscillatory error (0.0025%), and RK4 remains at $\sim 10^{-11}\%$. Euler is neither symplectic nor time-reversible; Velocity-Verlet preserves bounded energy behaviour as expected of a symplectic method; RK4 achieves the smallest short-time error but without the structural guarantees needed for long-time Hamiltonian simulation. Of the three, RK4 is the strongest verification reference on this problem, while Velocity-Verlet is the most suitable production integrator for MD.

3.2 Lennard–Jones Argon Fluid

The Lennard–Jones argon system ($N = 864$, $\rho = 0.8\sigma^{-3}$, $T_{\text{target}} = 94.4\text{ K}$, $\Delta t = 10^{-14}\text{ s}$) provides the first interacting many-body test of the solver. Following the startup procedure described in Section 2.5, the comparison below begins at the first frame of the post-startup NVE segment: in the saved production trajec-

Table 1. Summary of the 100-step (1 ps) post-startup NVE comparison.

Integrator	$\langle T \rangle \pm \sigma_T$ [K]	Max $ \Delta E/E_0 $ [%]	Mean $ \Delta E/E_0 $ [%]
Velocity-Verlet	94.42 ± 1.43	0.082	0.033
Forward Euler	185.27 ± 74.57	127.587	39.788

tories, step 0 is already the production initial frame after the final temperature rescale. The startup phase is therefore excluded from Figs. 4 and 5, so that the comparison reflects only the subsequent unthermostatted NVE evolution under each integrator. Because the initial FCC lattice is still structurally relaxing into a disordered fluid, the system is subject to large transient intermolecular forces during this phase.

Over this 1 ps window, the contrast between the two schemes is already unambiguous. Table 1 and Figure 4 detail the signed relative total-energy deviation $\Delta E/E_0$. In his foundational 1964 study, Rahman [5] noted that the total energy of his simulated argon system exhibited fluctuations of approximately 3 parts in 10^4 (0.03%). Our Velocity-Verlet implementation demonstrates good agreement with this baseline, maintaining a bounded oscillation with a maximum relative deviation of 0.082% (8.2 parts in 10^4). The slight increase in fluctuation amplitude relative to Rahman is consistent with our use of an unshifted hard cutoff at $r_c = 2.25\sigma$, which introduces minor energetic discontinuities when particle pairs cross the cutoff. Forward Euler, by contrast, exhibits rapid secular energy growth, reaching 127.6% by the end of the interval. Because both trajectories enter this NVE segment from the same nominal state after the final rescale, the subsequent divergence is strictly numerical. The non-symplectic Forward Euler update systematically distorts the phase-space volume, whereas Velocity-Verlet

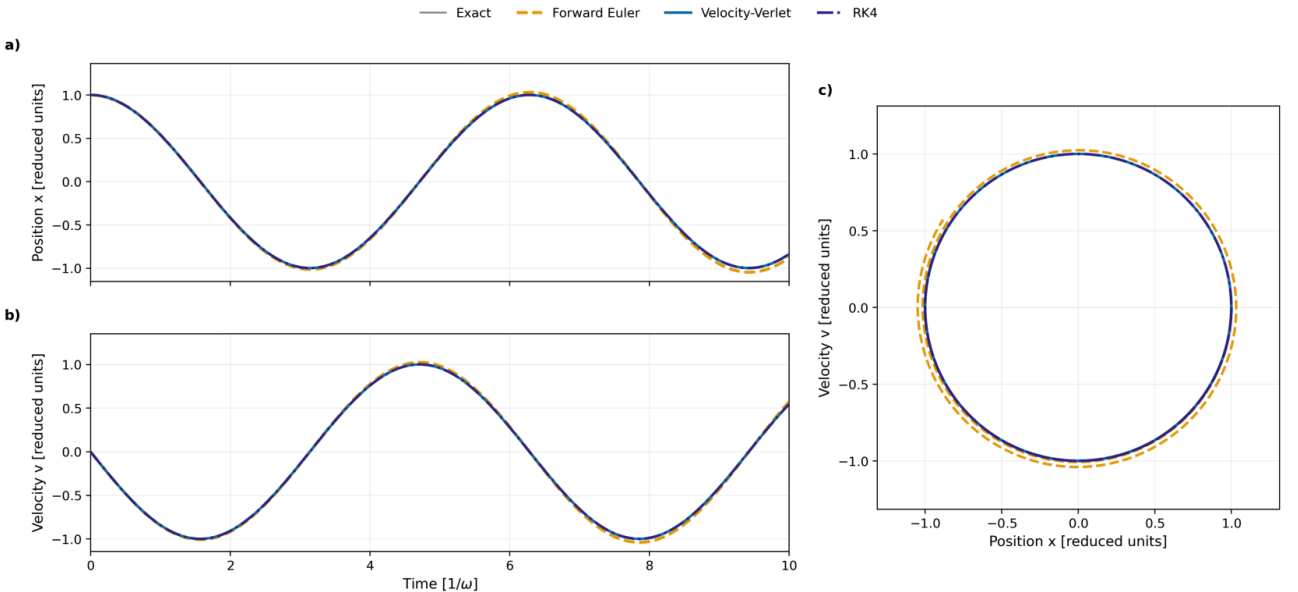


Figure 1. Harmonic-oscillator verification at $\Delta t = 0.01$ over $0 \leq t \leq 10$. (a) Position trajectory. (b) Velocity trajectory. (c) Phase-space portrait. All three schemes track the exact time traces closely, but the phase-space portrait discriminates: Euler spirals outward, Velocity-Verlet remains near-closed, and RK4 is visually indistinguishable from the exact orbit.

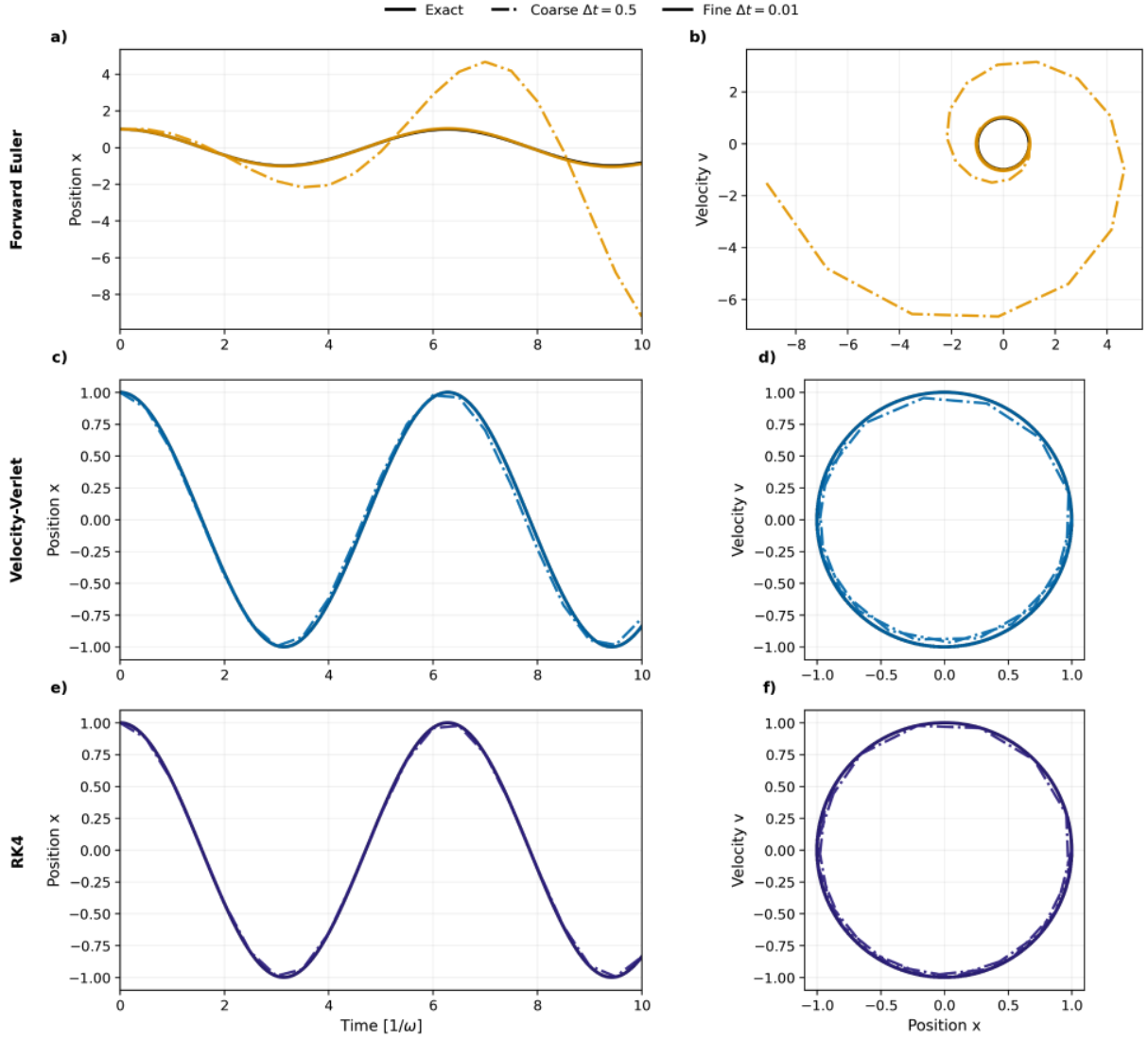


Figure 2. Coarse-versus-fine timestep comparison for the harmonic oscillator. Solid curves use $\Delta t = 0.01$; dash-dotted curves use $\Delta t = 0.5$. Euler becomes unstable at the coarse step; Velocity-Verlet remains bounded but inaccurate; RK4 remains accurate on this interval.

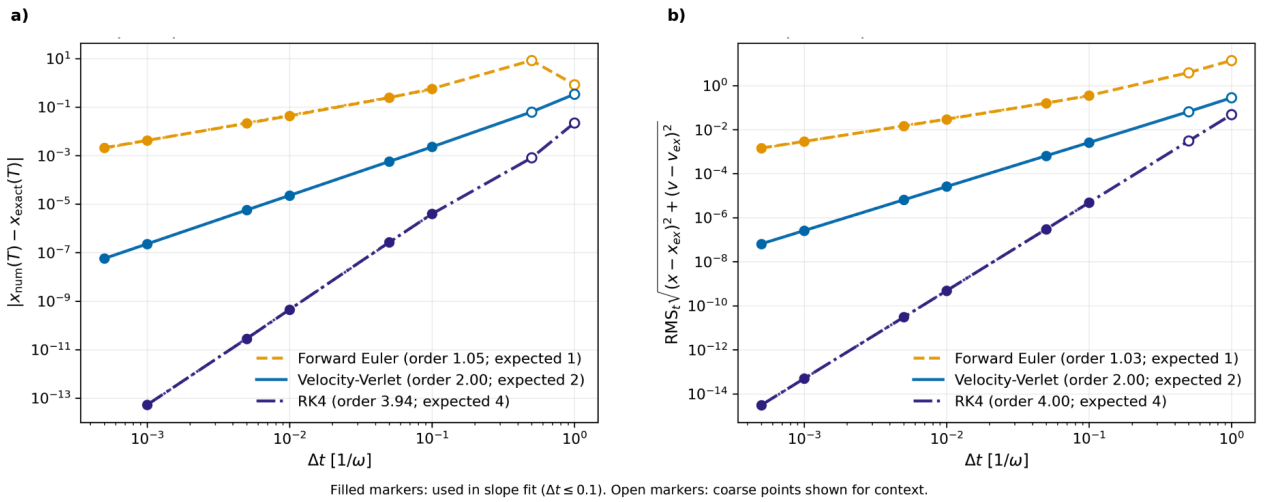


Figure 3. Convergence of harmonic-oscillator error with timestep. (a) Endpoint position error at $T = 10$. (b) RMS phase-space error over $0 \leq t \leq 10$. Filled markers denote the fitted range $\Delta t \leq 0.1$; open markers are coarser points shown for context. Fitted slopes recover the expected global orders 1, 2, and 4 for Euler, Velocity-Verlet, and RK4.

conserves a nearby shadow Hamiltonian.

The temperature traces in Figure 5 further highlight

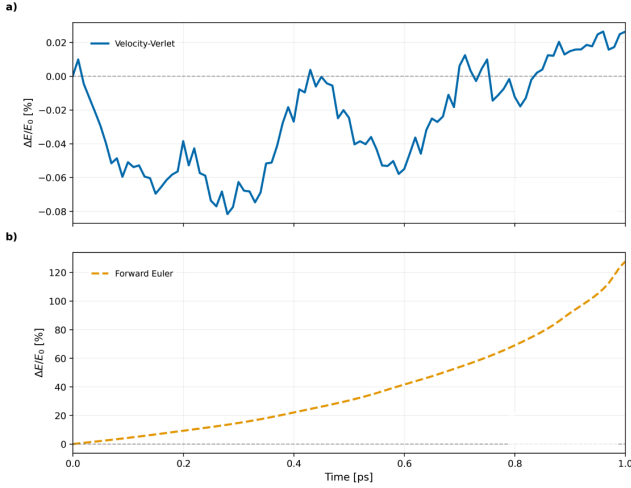


Figure 4. Signed relative total-energy deviation during the 100-step post-startup NVE comparison. **(a)** Velocity-Verlet: bounded oscillatory error with $\max |\Delta E/E_0| = 0.082\%$. **(b)** Forward Euler: strong secular drift to 127.6%.

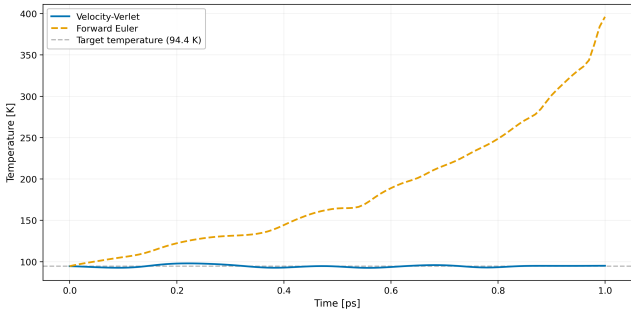


Figure 5. Instantaneous temperature during the same 100-step post-startup NVE comparison. Velocity-Verlet remains close to the 94.4 K reference line, whereas Euler undergoes rapid heating to 396.05 K.

this thermodynamic instability. Rahman’s simulation successfully equilibrated at 94.4 K. Our Velocity-Verlet trajectory remains highly stable around this intended state point ($\langle T \rangle = 94.42$ K and $\sigma_T = 1.43$ K), whereas Forward Euler heats continuously to 396.05 K within 1 ps, fundamentally failing to preserve the microcanonical ensemble.

While the 100-step interval demonstrates short-term thermodynamic stability, macroscopic structural observables cannot be statistically resolved over 10^{-12} s. To validate the structural fidelity of the solver against Rahman’s primary findings, an extended 20,000-step (20 ps) Velocity-Verlet production run was performed. The resulting radial distribution function $g(r)$ reproduces the standard liquid-argon shell structure: a first peak at $r/\sigma = 1.090$ (3.71 Å) with $g(r) = 2.838$, a first minimum at $r/\sigma = 1.550$ (5.27 Å) with $g(r) = 0.623$, a second peak at $r/\sigma = 2.070$ (7.04 Å) with $g(r) = 1.253$, and a long-range mean $\langle g(r) \rangle = 1.002$ for $r > 4\sigma$. This damped oscillatory form is the expected structural fingerprint of a dense liquid.

To ensure rigorous validation against the primary literature, quantitative comparisons in the text are restricted to the three radial coordinates explicitly annotated in Fig. 2 of Rahman [5] (3.7 Å, 7.0 Å and 10.4 Å).

Table 2. Comparison with the three radii explicitly annotated in Rahman’s Fig. 2. Distances are reported in Å.

Feature	Rahman r [Å]	Present r [Å]	$ \Delta r $ [Å]
First peak	3.7	3.71	0.01
2nd-shell peak	7.0	7.04	0.04
3rd-shell feature	10.4	10.40	0.00

Consequently, these structural benchmarks are reported in Å, whereas Figure 6 illustrates the broader, qualitative morphological agreement of the radial distribution function.

At the three annotated radii, the shell-location differences are no larger than 0.04 Å (Table 2). This is strong agreement given that Rahman’s figure provides graphical rather than tabulated RDF information. The remaining comparison is therefore best read from the overall shell structure and long-range damping in Figure 6.

Overall, the post-startup 1 ps NVE comparison shows that Velocity-Verlet remains stable over the strictly mandated window, successfully mirroring the energy-conservation behaviour reported by Rahman [5], whereas Euler injects enough energy to drive the system rapidly out of the intended regime. The longer Velocity-Verlet run then reproduces the expected liquid-argon shell structure at the Rahman state point.

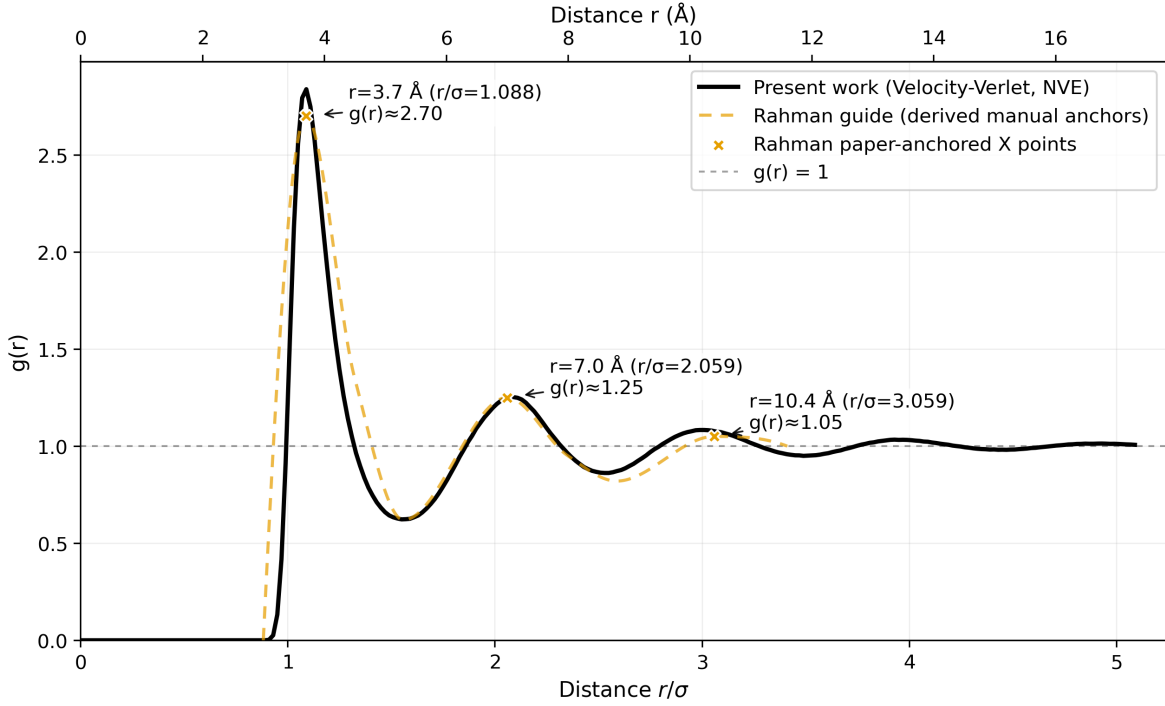


Figure 6. Radial distribution function from a separate 20 ps Velocity-Verlet NVE run ($N = 864$), compared with an overlay guide based on Fig. 2 of Rahman [5]. The radii 3.7 Å, 7.0 Å and 10.4 Å are explicitly annotated in the paper; the remaining dashed-guide anchors and all Rahman ordinates are graphical estimates from the printed curve. The comparison should therefore be read as qualitative / semi-quantitative, strongest in shell locations and overall liquid-like shape rather than exact ordinate-by-ordinate agreement.

3.3 Parallel Scaling Analysis

All timing runs use Velocity-Verlet with fixed-seed identical initial conditions, $\Delta t = 10^{-14}$ s, and no file output. Each configuration is repeated 11 times; reported timings are medians over these repetitions. Strong scaling is measured at fixed $N = 2048$ using 500 timed steps; problem-size scaling at fixed $P = 16$ using 2000 timed steps. The longer step count for problem-size scaling ensures that small- N cases are not dominated by timer granularity or startup overhead.

Prior to performance profiling, the parallel logic was verified by comparing unthermostatted energy trajectories across varying process counts. Trajectories generated with identical initial seeds yielded bitwise agreement between $P = 1$ and $P = 2$. Discrepancies at larger P remained strictly within the bounds expected from floating-point non-associativity in MPI reduction trees, confirming that subsequent speedup metrics reflect true parallel work distribution rather than altered dynamics.

3.3.1 Strong scaling

At fixed $N = 2048$, the strong-scaling results in Fig. 7 show near-ideal speedup at moderate process counts: $S(8) = 7.55$ with $E(8) = 0.944$. Even at $P = 32$ the code achieves $S(32) = 24.12$ with $E(32) = 0.754$, which is strong performance for a direct-force code requiring one global position exchange per Velocity-Verlet step.

The systematic efficiency loss is communication-driven. As P increases, the $\mathcal{O}(N^2/P)$ local arithmetic work falls while the cost of reconstructing the replicated

position array via `MPI_Allgather` persists. The measured critical-path communication share accordingly rises from 1.1% at $P = 2$ to 15.2% at $P = 32$ (Fig. 7c). This matches the behaviour expected when message-passing overhead is included in a simple performance model: $t_P \approx t_s + t_p/P + c\lambda P$, where the communication term $c\lambda P$ becomes an increasingly large fraction of the total runtime as the work per rank shrinks, even when the code is still scaling reasonably well. The standard Amdahl fit (Eq. 21) with $f \approx 0.0103$ provides a compact empirical description of the observed curve, but f should not be interpreted as a literal machine-independent serial fraction: at fixed $N = 2048$ and $P = 32$, each rank owns only 64 particles, so the problem is fundamentally starved of arithmetic relative to communication latency. In conceptual terms, a weak-scaling-style increase of problem size with process count would be expected to sustain efficiency better than fixed- N strong scaling for this direct-summation code, because the $\mathcal{O}(N^2/P)$ work grows faster than the $\mathcal{O}(N)$ communication volume.

3.3.2 Problem-size scaling

The fixed- P problem-size-scaling results in Fig. 8 confirm the expected cost structure of direct all-pairs force evaluation. Over the fit domain $N \geq 500$, where the quadratic force cost dominates, the measured wall time scales as $N^{1.85}$ and the compute-only time (wall minus critical-path communication) as $N^{1.93}$. The compute-only exponent is the more algorithmically informative quantity: it isolates the force-evaluation cost and lies close to the expected $\mathcal{O}(N^2)$. The slightly sub-

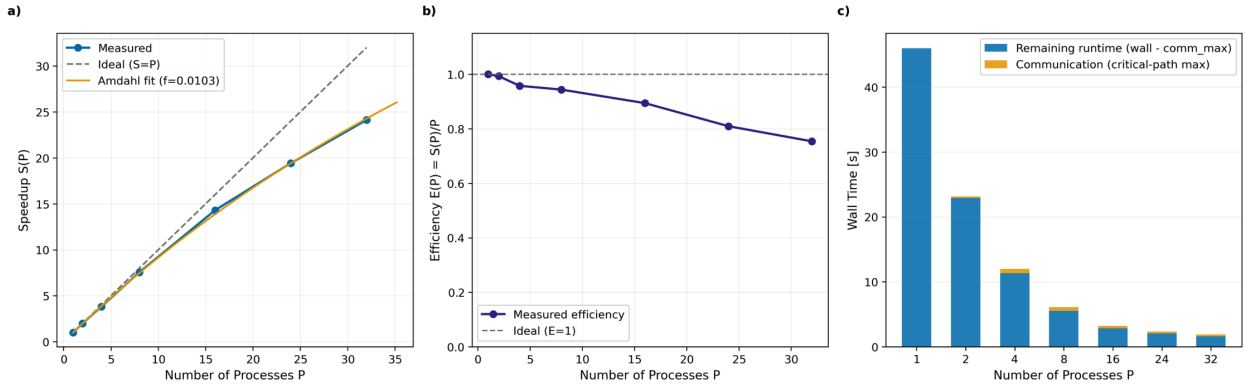


Figure 7. Strong scaling at fixed $N = 2048$ (500 timed steps per run, median of 11 repetitions). (a) Speedup with ideal line and Amdahl fit ($f = 0.0103$). (b) Parallel efficiency. (c) Breakdown into critical-path communication and remaining wall time, showing the growing communication share as work per rank decreases.

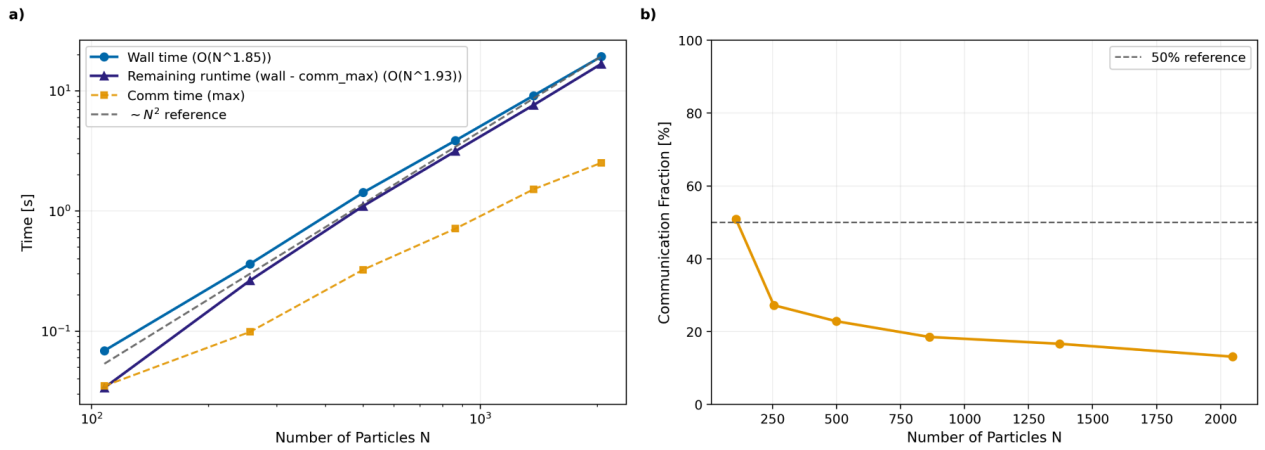


Figure 8. Problem-size scaling at fixed $P = 16$ (2000 timed steps per run, median of 11 repetitions). (a) Wall time and wall-minus-communication runtime versus N on log-log axes, with an N^2 reference line. Power-law fits are over $N \geq 500$. (b) Communication fraction versus N .

quadratic wall-time exponent reflects the inclusion of the communication term, whose relative weight diminishes with N . This is directly visible in the communication fraction (Fig. 8b), which falls from 50.8% at $N = 108$ to 13.1% at $N = 2048$: at small N the problem is communication-dominated, while the quadratic force cost increasingly controls the total runtime as N grows. The observed trend is one of convergence toward the expected $O(N^2)$ algorithmic cost at larger system sizes—precisely the regime in which parallelisation provides the greatest benefit, since the arithmetic workload grows faster than the communication overhead.

Both scaling experiments are consistent with the parallel design described in the Methodology: each rank computes forces for its owned particles against a globally reconstructed position array obtained through `MPI_Allgatherv`. Velocity-Verlet and Euler require one such collective per step, RK4 would require four. The measured behaviour—good acceleration at modest P , near-quadratic growth with N , and increasingly visible communication limits under strong scaling—follows directly from this design.

4 Conclusions

This project shows that, in molecular dynamics, formal order alone does not determine practical suitability.

The harmonic-oscillator tests recover the expected global convergence orders for Euler, Velocity-Verlet, and RK4, confirming correct implementation, but they also show the more important structural distinction: Euler produces secular energy drift and phase-space distortion, whereas Velocity-Verlet retains the bounded energy behaviour expected of a symplectic method. RK4 is highly accurate on short horizons, reaching the double-precision floor in the verification tests, but that does not make it the best method for long-time MD.

The Lennard-Jones argon results show that this distinction carries over directly to the interacting many-body system. In the required 100-step, 1 ps NVE run, Velocity-Verlet remains stable, with mean temperature 94.42 K and maximum relative energy deviation 0.082%, whereas Euler exhibits strong energy drift and heats to 396.05 K. This thermal runaway explicitly violates the microcanonical (NVE) ensemble constraints, disqualifying Euler as a viable production integrator for thermodynamic sampling. RK4 should also not be preferred here: although more accurate in the harmonic-oscillator benchmark, it is non-symplectic and, in the present MPI implementation, would require four globally consistent force stages per timestep. Velocity-Verlet is therefore the correct production choice in this

code, combining bounded long-time behaviour with the lowest communication cost among the tested interacting schemes.

The longer Velocity-Verlet RDF run further supports the physical plausibility of the implementation. The computed $g(r)$ reproduces the expected liquid shell structure and broadly agrees with the Rahman reference, but the comparison should remain qualitative rather than exact. Taken together, the harmonic-oscillator and LJ results show that the solver is numerically correct, physically credible, and that stable MD depends more on preserving Hamiltonian structure than on maximising formal order alone.

From the computational perspective, the MPI particle-decomposition strategy performs well but reveals its own limits. Strong scaling remains good to moderate process counts, reaching $S(32) = 24.12$ with efficiency 0.754, while fixed- P problem-size scaling remains close to the expected $\mathcal{O}(N^2)$ cost of direct pair forces. The loss of efficiency at larger P is increasingly communication-driven, since each timestep still requires global position reconstruction. In conceptual weak-scaling terms, increasing problem size with process count would be expected to sustain efficiency better than the fixed- N strong-scaling case studied here, because the local arithmetic grows faster than the replicated-position communication volume. The present AoS data layout is also not maximally SIMD-friendly, so the measured strong-scaling efficiencies should be read in light of a deliberately simple rather than fully performance-tuned single-node implementation. For practical large-scale MD, this points directly to the standard production improvements: neighbour or cell lists to avoid direct all-pairs cost, spatial domain decomposition to localise communication, and smoother cutoff treatments to reduce cutoff artefacts. The present implementation is therefore best understood not as a production endpoint, but as a clear and successful demonstration of the central MD trade-offs between numerical stability, physical fidelity, and parallel efficiency.

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Appendix

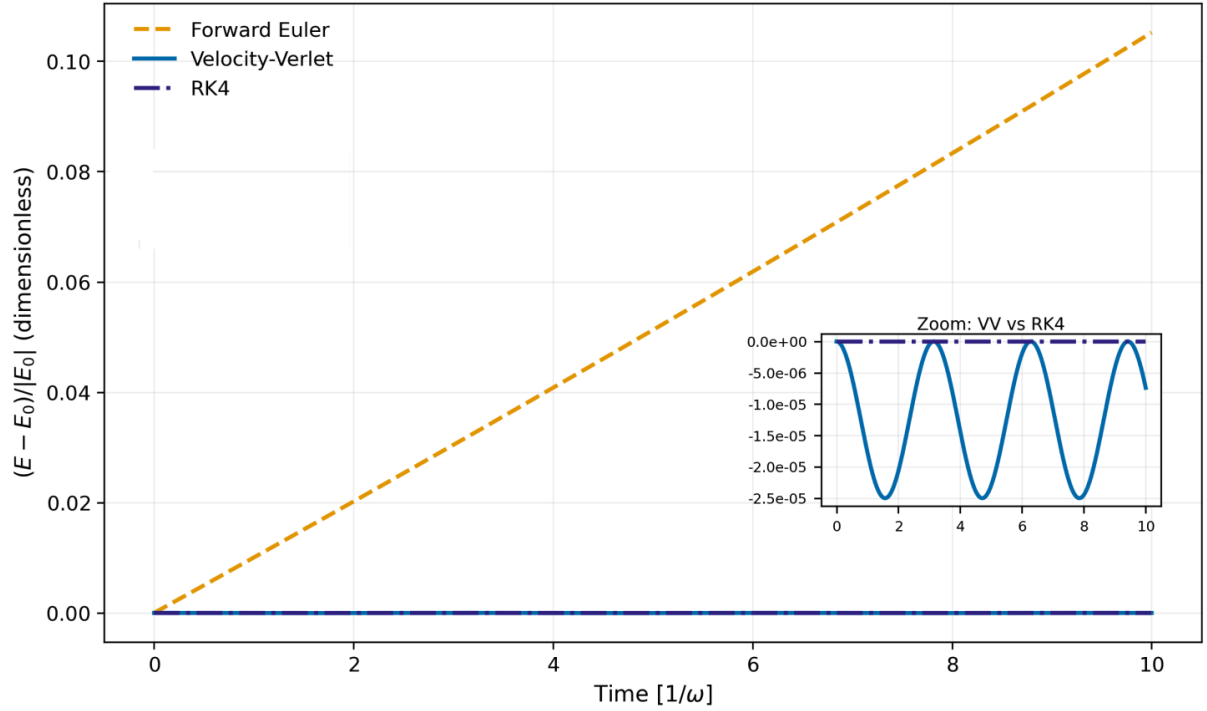


Figure 9. Supporting harmonic-oscillator energy diagnostic at $\Delta t = 0.01$. Euler shows clear secular drift over $0 \leq t \leq 10$. Velocity-Verlet shows a small bounded oscillatory energy error, consistent with the expected behaviour of a symplectic method at fixed timestep. RK4 remains extremely accurate on this interval, but this should be read as short-horizon accuracy rather than as evidence of symplecticity.

Table 3. Representative harmonic-oscillator endpoint errors together with fitted global convergence orders. Coarser points are included for context, but only $\Delta t \leq 0.1$ is used for the fitted slopes reported in Fig. 3.

Δt	Euler	Velocity-Verlet	RK4
10^0	7.0×10^0	3.4×10^{-1}	2.2×10^{-2}
10^{-1}	2.3×10^{-1}	2.3×10^{-3}	3.9×10^{-6}
10^{-2}	2.0×10^{-2}	2.3×10^{-5}	4.5×10^{-10}
10^{-3}	2.0×10^{-3}	2.3×10^{-7}	5.3×10^{-14}
5×10^{-4}	9.8×10^{-4}	5.7×10^{-8}	3.6×10^{-15}
Fitted slope	1.05	2.00	3.94

Table 4. Guide points used for the Rahman overlay in Fig. 6. The table is divided into (i) radii explicitly annotated in Rahman’s Fig. 2 and (ii) auxiliary graphical anchors used only to sketch the dashed guide. Only the radii 3.7 Å, 7.0 Å and 10.4 Å are explicitly marked in the published figure; all ordinates and all auxiliary points are graphical estimates from the printed curve.

Category	r [Å]	r/σ	Guide $g(r)$	Note
<i>Paper-annotated radii</i>				
Annotated radius	3.7	1.088	2.70	Radius explicitly labelled in Rahman’s Fig. 2; ordinate estimated graphically from the printed first peak.
Annotated radius	7.0	2.059	1.25	Radius explicitly labelled in Rahman’s Fig. 2; ordinate estimated graphically from the printed second-shell peak.
Annotated radius	10.4	3.059	1.05	Radius explicitly labelled in Rahman’s Fig. 2; ordinate estimated graphically from the printed third-shell oscillation.
<i>Auxiliary graphical anchors for the dashed guide</i>				
Guide anchor	3.0	0.882	0.00	Onset / exclusion-region anchor.
Guide anchor	4.5	1.324	1.35	Descending shoulder after the first peak.
Guide anchor	5.3	1.559	0.62	First minimum.
Guide anchor	8.0	2.353	0.95	Post-second-peak descent.
Guide anchor	8.8	2.588	0.82	Oscillatory trough / second minimum.
Guide anchor	11.6	3.412	1.00	Long-range return toward $g(r) = 1$.